

HOSPITAL OPERATIONS & PATIENT ANALYTICS DASHBOARD

Infosys Springboard Internship Project



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Project Overview ☐ ☒

About the Project

Transform hospital data into interactive dashboards for better decision-making.

Focus Areas

- Patient Admissions
- Revenue Analysis
- Bed Occupancy
- Department Performance
- Resource Utilization
- Emergency Cases

Outcome

4 Interactive Dashboards developed using Tableau and Power BI.





PROJECT WORKFLOW



Internship Journey.

*Data Collection
& Data Cleaning*

KPI Development

Dashboard Storyboarding

Dashboard Development

Testing & Validation





DATA COLLECTION & CLEANING

Module 1 & Module 2

Data Collection

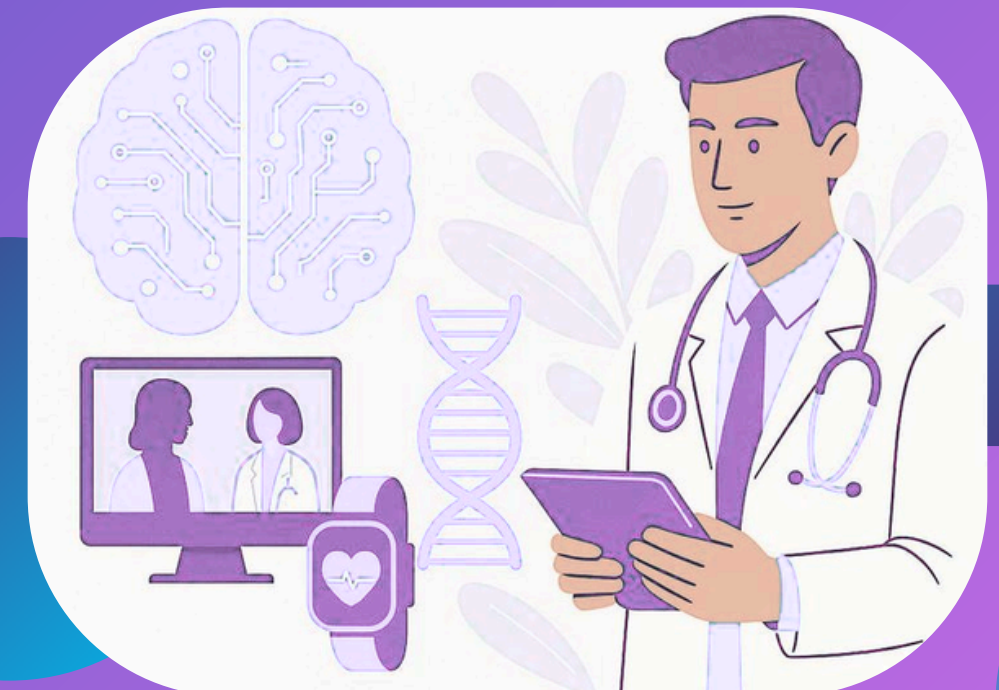
- Healthcare datasets collected from Kaggle
- Hospital Operations Dataset
- Patient Admission Dataset

Data Cleaning

- Missing values handled
- Duplicate records removed
- Data formatting standardized
- Dataset validation performed

Result

Clean and analysis-ready dataset





Project Overview

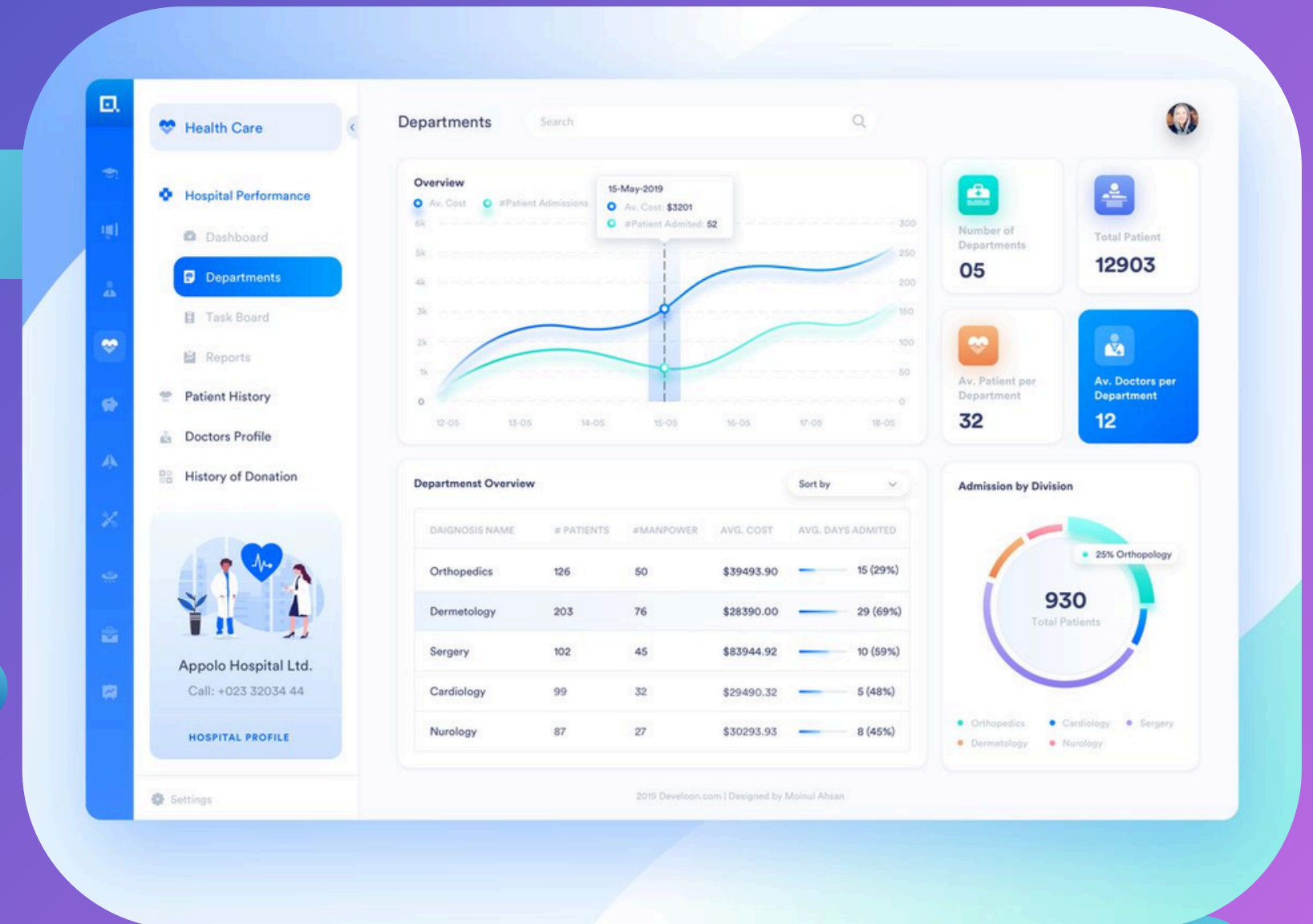


Key Performance Indicators (KPIs)

- Total Patients
- Total Revenue
- Bed Occupancy Rate
- Average Length of Stay
- Department-wise Patient Count
- Resource Utilization %
- Doctor Utilization %
- Emergency Cases

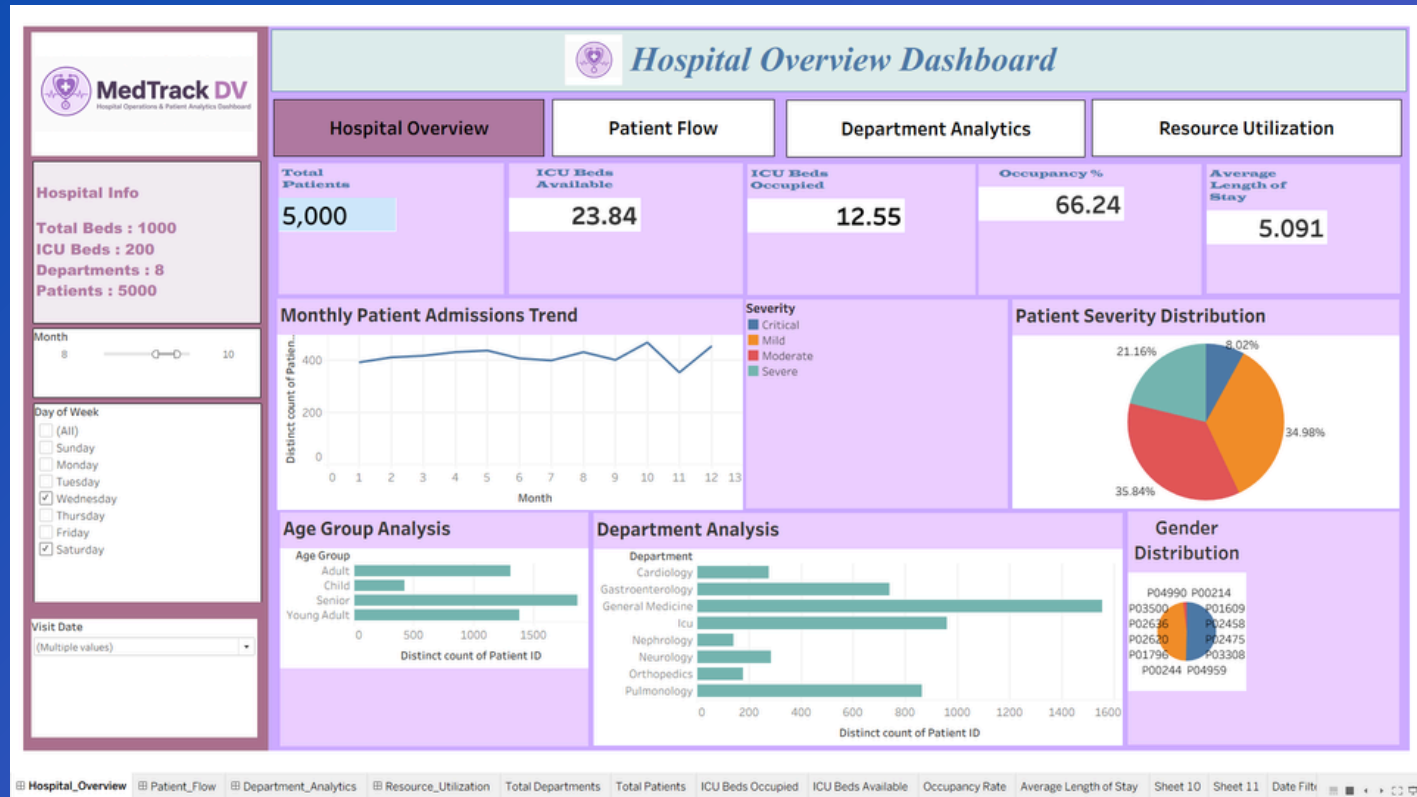
Purpose

Measure hospital performance and operational efficiency.





DASHBOARDS CREATED



Dashboard 1



Hospital Overview

Dashboard 2



Patient Flow Analysis

Dashboard 3

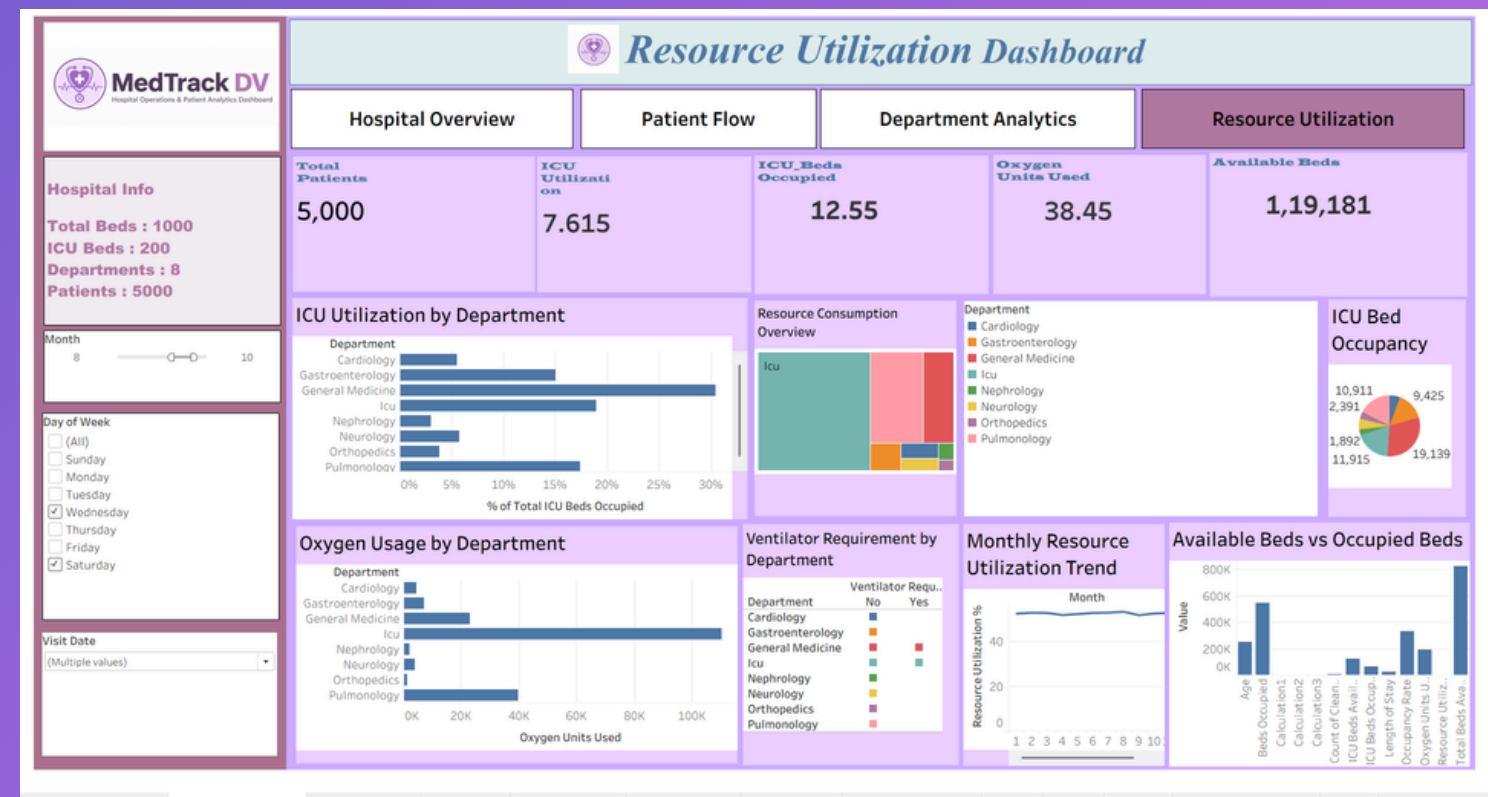
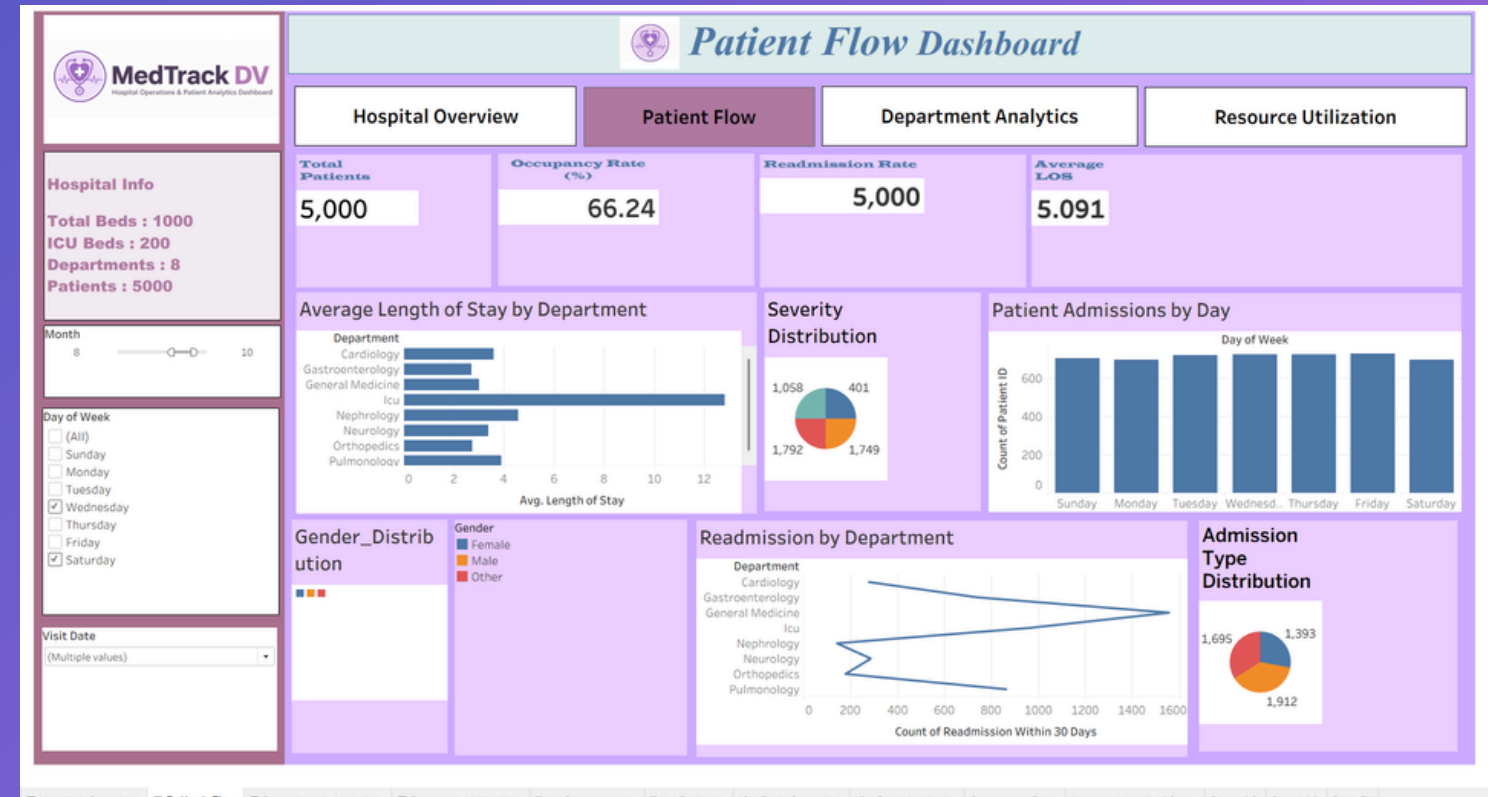
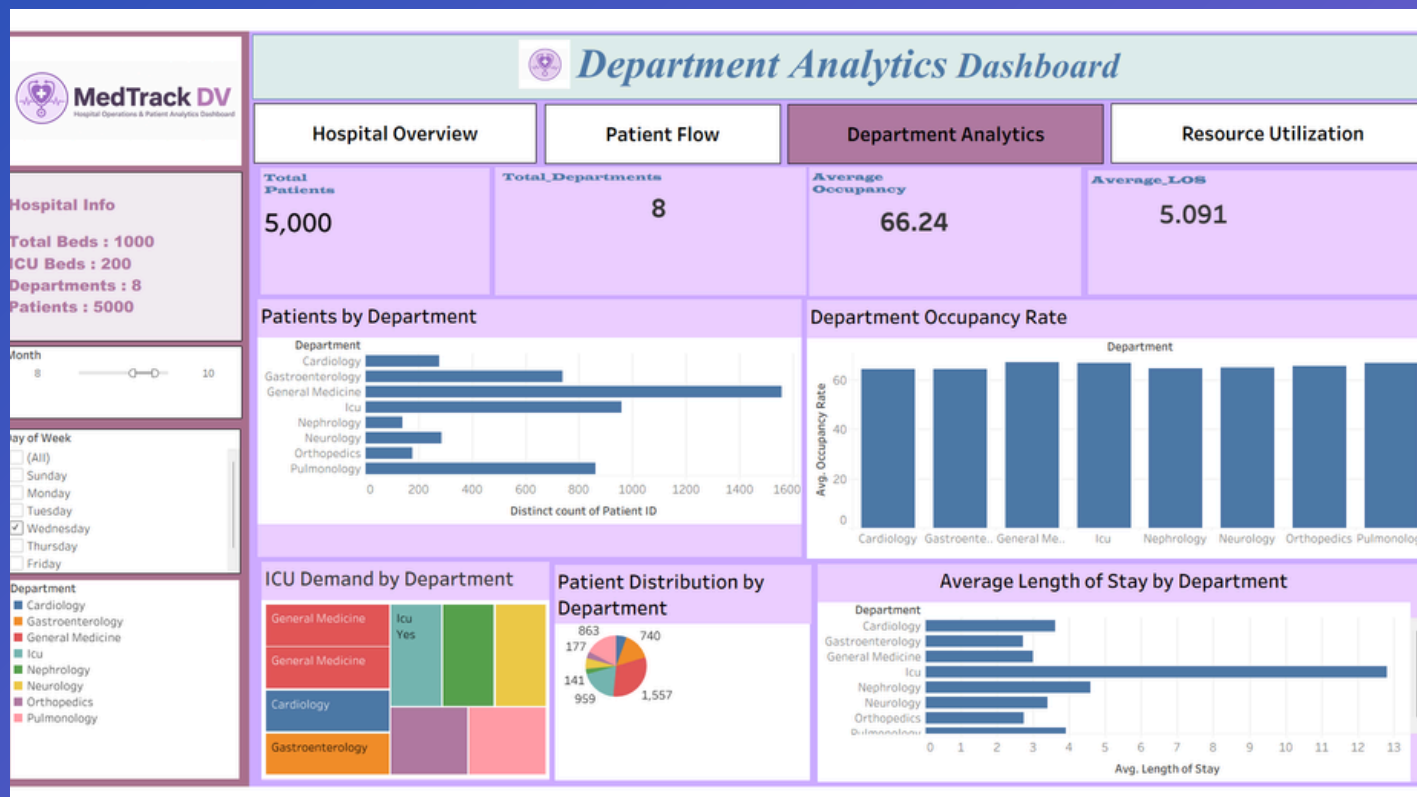


Department Analytics

Dashboard 4



Resource Utilization





DASHBOARD FEATURES

Hospital Overview

- Revenue Monitoring
- Bed Occupancy
- Emergency Cases

Patient Flow

- Admissions
- Discharges
- Length of Stay

Department Analytics

- Department Comparison
- Patient Distribution

Resource Utilization

- Doctor Utilization
- Resource Efficiency
- Capacity Monitoring



TOOLS & TECHNOLOGIES



Kaggle

Dataset collection and healthcare data sourcing



Excel

Data inspection, validation, and preliminary analysis



Python

Data cleaning, preprocessing, and transformation



Tableau

Interactive dashboard design and visualization



Power BI

Business intelligence reporting and dashboard creation



GitHub

Version control and project documentation management



GitHub Desktop

Repository synchronization and project uploads



TOOLS & TECHNOLOGIES

Challenges

- *Data Cleaning*
- *KPI Validation*
- *Dashboard Design*
- *Data Consistency*
- *GitHub Repository Management*

Key Learnings

- ✓ *Healthcare Analytics*
- ✓ *Data Visualization*
- ✓ *Tableau*
- ✓ *Power BI*
- ✓ *GitHub Workflow*
- ✓ *Business Intelligence Reporting*





CONCLUSION

Project Outcome

Successfully developed a Hospital Operations & Patient Analytics Dashboard that transforms raw healthcare data into meaningful insights.

Deliverables

- ✓ 4 Interactive Dashboards
- ✓ Tableau & Power BI Reports
- ✓ GitHub Repository
- ✓ Final Documentation

Thank You

